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Misallocation and Capital Market Integration: Evidence from India

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1 Big picture

- **Question.** Does opening a developing country to foreign equity capital reduce within-industry misallocation, or does it simply increase the aggregate capital stock without improving allocation?
- **Setting.** India's staggered liberalization of foreign direct investment at the 5-digit manufacturing-industry level. The key post-2000 liberalization episodes occur in 2001 and 2006, after the main early-1990s reform wave.
- **Core empirical idea.** If foreign capital reduces capital market frictions, firms that had *ex ante* high marginal revenue products of capital should expand more after the reform than firms that had low MRPK. That is exactly the pattern the paper estimates.
- **Main micro result.** Relative to low-MRPK firms, high-MRPK firms in treated industries raise revenues by about 23%, physical capital by about 53%, and wage bills by about 28%, while their MRPK falls by about 33%.
- **Interpretation.** The reform does not merely raise average capital. It reallocates capital toward firms with higher wedges / higher marginal returns, which is the hallmark of a reduction in misallocation.
- **Mechanism.** The effects are much stronger in states with less developed local banking sectors, suggesting that poor domestic credit intermediation is one source of the original misallocation.
- **Broader result.** The paper also finds evidence of lower labor misallocation, lower product prices for initially constrained firms, and an increase in treated industries' aggregate productivity as measured by the Solow residual.
- **Why the paper matters.** The contribution is methodological as well as substantive: the authors show how a natural experiment can identify *changes* in wedges and can be used to bound aggregate productivity effects without leaning on the usual cross-sectional assumptions that often inflate measured misallocation.

2 Data and measurement (institutional setting and key objects)

2.1 Foreign-capital liberalization in India

- **Institutional background.** Before 1991, foreign investment in most industries required case-by-case approval under the Foreign Exchange Regulation Act, and foreign ownership caps were generally below 40%.
- **Policy shock studied here.** The paper focuses on later manufacturing reforms in the 2000s that granted automatic approval and raised foreign-equity caps to at least 51% in selected industries.
- **Why these episodes are useful.** The policy varies across narrowly defined industries and across time, creating quasi-experimental within-country variation that is much cleaner than comparing countries or broad macro periods.
- **Treatment timing.** After dropping a very small 1998 liberalization, the relevant post-1995 liberalization episodes occur in 2001 and 2006.

- **Treatment intensity in the sample.** Around 10% of manufacturing firms in the main estimation sample are in industries affected by the post-2000 liberalizations.

2.2 Main firm-level data: *Prowess*

- **Source.** CMIE’s *Prowess* database, a panel of large and medium-sized Indian firms, including all public firms and many private firms.
- **Coverage.** The paper states that *Prowess* covers more than 70% of organized-sector economic activity and about 75% of corporate tax collections in India.
- **Study period.** 1995–2015. The authors start in 1995 to avoid the noisier early years of *Prowess* and to stay away from the major 1991 reform wave.
- **Final main sample.** 5,013 distinct firms, 337 distinct 5-digit industries, and 63,149 firm-year observations after the sample restrictions needed for the misallocation design.
- **Variables observed.** Sales, physical assets, wage bills, material inputs, ownership, age, and (crucially) product-level quantity and sales information for many firms.

2.3 Product and supplementary data

- **Product-level data.** *Prowess* reports firm-product sales and quantities, so the authors can construct unit prices and output at the product level.
- **First-stage datasets.** They supplement the main data with RBI foreign-loan records and CMIE *CapEx* data on large capital projects to show that treated firms gain foreign financing.
- **Representativeness check.** They use the Annual Survey of Industries (*ASI*) to verify that the decline in MRPK dispersion also appears in a representative sample of formal manufacturing establishments.
- **Mechanism data.** State-level banking credit from the Reserve Bank of India is used to proxy local financial development before the reform.
- **Robustness data.** The paper also controls for dereservation reforms and trade liberalization using industry-level policy data.

2.4 Key measurement choices

- **MRPK.** Under a Cobb-Douglas revenue production function, $MRPK_{ijt} = \alpha_j^k \frac{Revenue_{ijt}}{K_{ijt}}$, so $Revenue_{ijt}/K_{ijt}$ is the within-industry proxy for the marginal revenue product of capital.
- **High-MRPK classification.** A firm is high-MRPK if its average $Revenue/K$ over 1995–2000 is above the 4-digit industry median.
- **MRPL.** The labor analogue uses $Revenue/L$, where labor is proxied by the wage bill because headcount is unavailable in *Prowess*.
- **TFPQ.** The authors estimate revenue production functions using Levinsohn-Petrin with Wooldridge’s GMM implementation, then subtract log average product prices from TFPR: $\log TFPQ = \log TFPR - \log \tilde{p}$.

- **Sample for TFPQ.** The TFPQ analysis is smaller (43,791 firm-year observations), since it requires both full input data and product-price data.
- **Local financial development.** Measured as log average total bank credit in a firm’s state over 1995–2000.

2.5 Descriptive statistics

- **Treatment versus control.** Table I shows that treated firms are not obviously larger on every dimension than controls, but they do have slightly higher mean *Revenue/K* before the reform (9.3 versus 7.5).
- **Ownership structure.** Most firms are domestic; foreign-owned and state-owned firms are a minority in both treated and untreated industries.
- **Interpretation.** The paper is fundamentally about the formal corporate sector, not the universe of informal firms. That is appropriate for the Solow-residual exercise, since large firms carry most sales weight.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Wedges on inputs and the firm’s first-order conditions

The paper models misallocation as wedges on the prices of inputs $x \in \{K, L, M\}$. The allocative price paid by firm i is $(1 + \tilde{\tau}_i^x)p^x$, so the profit function of a single-product firm is

$$\pi_i = p_i f_i(K_i, L_i, M_i) - \sum_{x \in \{K, L, M\}} (1 + \tilde{\tau}_i^x) p^x x_i. \quad (1)$$

Cost minimization implies

$$p_i \frac{\partial f_i(K_i, L_i, M_i)}{\partial x_i} = \mu_i (1 + \tilde{\tau}_i^x) p^x, \quad (2)$$

where μ_i is the markup (the output wedge). Defining the combined wedge as $1 + \tau_i^x = \mu_i (1 + \tilde{\tau}_i^x)$, higher wedges imply higher marginal revenue products of the relevant input.

- **Interpretation.** Dispersion in marginal revenue products is informative about wedges: a firm with a high observed MRPK is a firm that is either taxed more heavily on capital or otherwise more constrained in obtaining capital.
- **Why this matters.** The empirical design is built around the idea that if the reform relaxes capital-market wedges, capital should expand most for firms with the highest *ex ante* MRPK.

3.2 Solow residual and aggregate productivity

The paper measures aggregate productivity effects through the Solow residual of the set I of treated industries. First,

$$\Delta Solow_I = \Delta Net\ Output_I - \Delta Net\ Input_I. \quad (3)$$

Let $c_i = y_i - \sum_{s \in I} y_{si}$ denote net output of good i , and $C_I = \sum_{i \in I} p_i c_i$ denote treated industries' net output valued at fixed prices. Then the discrete-time Solow residual is

$$\Delta Solow_I = \Delta \log C_I - \sum_{s \notin I} \frac{\sum_{i \in I} p_s y_{is}}{\sum_{i \in I} p_i c_i} \Delta \log \left(\sum_{i \in I} y_{is} \right). \quad (4)$$

Using results from Petrin-Levinsohn and Baqaee-Farhi, the first-order approximation for the change in the Solow residual is

$$\Delta Solow_{I,t} \approx \sum_{i \in I} \lambda_i \Delta \log A_i + \sum_{i \in I} \sum_{x \in \{K, L, M\}} \lambda_i \alpha_i^x \frac{\tau_i^x}{1 + \tau_i^x} \Delta \log x_i. \quad (5)$$

- **Interpretation of λ_i .** λ_i is firm i 's sales share in treated industries' net output, so larger firms receive more aggregate weight.
- **Interpretation of the reallocation term.** The Solow residual rises if inputs grow more for firms with larger positive wedges, that is, for firms with higher marginal revenue products.
- **Big conceptual payoff.** This formula shows why the paper can detect reductions in misallocation even when total capital rises. Reallocation is beneficial as long as additional inputs disproportionately flow to the firms with high wedges.
- **Advantage over Hsieh-Klenow variance methods.** The paper emphasizes that this approach does *not* require constant returns, CES aggregation, or joint lognormality of TFPR and TFPQ.

3.3 Revenue production function and measurement of MRPK

The paper assumes Cobb-Douglas revenue production functions:

$$Revenue_{ijt} = TFPR_{ijt} K_{ijt}^{\alpha_j^k} L_{ijt}^{\alpha_j^\ell} M_{ijt}^{\alpha_j^m}. \quad (6)$$

Under this specification,

$$MRPK_{ijt} = \frac{\partial Revenue_{ijt}}{\partial K_{ijt}} = \alpha_j^k \frac{Revenue_{ijt}}{K_{ijt}}. \quad (7)$$

- **Interpretation.** Within an industry, $Revenue/K$ is proportional to MRPK if firms share the same output elasticity α_j^k .
- **Empirical use.** The paper averages $Revenue/K$ over 1995–2000, then splits firms into high and low MRPK bins at the 4-digit industry median.
- **TFPQ measurement.** The paper separately estimates production-function coefficients and uses product prices to move from TFPR to TFPQ.

3.4 Main heterogeneous-effects difference-in-differences specification

The core estimating equation is

$$Outcome_{ijt} = \beta_1 Reform_{jt} + \beta_2 (Reform_{jt} \times I_i^{HighMRPK}) + \mathbf{X}_{it} + \theta_i + \delta_t + \varepsilon_{ijt}. \quad (8)$$

- **Dependent variable.** In different regressions, $Outcome_{ijt}$ is log revenues, log capital, log wages, or log MRPK.
- **Treatment.** $Reform_{jt} = 1$ once industry j is liberalized.
- **Heterogeneity.** $I_i^{HighMRPK} = 1$ if firm i had above-median pre-reform MRPK.
- **Controls.** $\mathbf{X}_{it}$ includes firm age and pre-treatment size-by-year fixed effects. The regressions also include firm fixed effects and year fixed effects.
- **Inference.** Standard errors are two-way clustered at the 4-digit industry and year level.
- **Interpretation of β_2 .** This is the key coefficient. If $\beta_2 > 0$ when the dependent variable is capital, capital rises more for ex ante high-wedge firms, which implies a decline in within-industry capital misallocation.
- **Interpretation of β_1 .** This is the treatment effect for low-MRPK firms; $\beta_1 + \beta_2$ is the total treatment effect for high-MRPK firms.

3.5 Aggregate decomposition based on firm-level inputs

When the authors move from reduced-form micro estimates to aggregate productivity, they focus on the input-reallocation component:

$$\Delta Solow_{I,t} = \sum_{i \in I} \sum_{x \in \{K,L,M\}} \lambda_i \alpha_i^x \frac{\tau_i^x}{1 + \tau_i^x} \Delta \log x_i. \quad (9)$$

They then discuss two ways of identifying the wedges τ_i^x .

- **Conventional approach.** Infer baseline wedges directly from pre-treatment cross-sectional expenditure shares:

$$\tau_i^K = \alpha_j^K \frac{p_i y_i}{r K_i} - 1, \quad \tau_i^L = \alpha_j^L \frac{p_i y_i}{w L_i} - 1.$$

- **Concern.** This cross-sectional method is sensitive to measurement error and model misspecification, and can overstate the gains from reallocation.

3.6 Lower-bound aggregation strategy: Assumption 1 and Proposition 1

To get a more conservative aggregate estimate, the paper imposes:

- if the pre-reform wedge is positive, the reform weakly moves it toward zero and the firm's input weakly rises;
- if the pre-reform wedge is negative, the reform weakly moves it toward zero and the firm's input weakly falls.

Under that assumption, the paper proves

$$\sum_{i,x} \lambda_i \alpha_i^x \frac{\tau_i^{x,pre}}{1 + \tau_i^{x,pre}} \Delta \log x_i \geq - \sum_{i,x} \lambda_i \alpha_i^x \frac{\Delta \tau_i^x}{1 - \Delta \tau_i^x} \Delta \log x_i. \quad (10)$$

- **Interpretation.** The right-hand side gives a lower bound for the first-order aggregate productivity effect of the policy.
- **Why it is useful.** If high-MRPK firms' measured post-reform dispersion partly reflects noise rather than true wedges, this method attributes that residual dispersion to mismeasurement instead of to genuine misallocation. So it is deliberately conservative.

3.7 Estimating changes in wedges directly from the natural experiment

To identify the change in wedges, the paper estimates flexible difference-in-differences regressions with marginal revenue products on the left-hand side. For capital, the generic example is

$$\log \text{MRPK}_{ijt} = g(\text{Reform}_{jt}, C_{it}) + \mathbf{X}_{it} + \theta_i + \delta_t + \varepsilon_{ijt}, \quad (11)$$

and then maps the fitted value into a change in wedges via $\Delta \hat{\tau}_i^k = e^{\hat{g}_i(1, C_{it})} - 1$.

- **Interpretation.** Instead of inferring wedges from cross-sectional levels, the authors identify *changes* in wedges caused by the policy.
- **Why more credible.** Firm fixed effects absorb time-invariant measurement error, and year effects absorb aggregate shocks. The identifying assumption is only that untreated industries are a valid counterfactual for treated industries.
- **Materials.** The aggregation exercise assumes materials are not misallocated, so $\tau_i^M = 0$.

4 Identification and empirical credibility

4.1 Main identifying assumption

- The design does *not* require industries to be randomly assigned to liberalization.
- It only requires that in the absence of the reform, high-MRPK firms would not have grown faster than low-MRPK firms in treated industries relative to untreated industries.
- This is a relative-trends assumption: the difference between high- and low-MRPK firms inside treated industries should have followed the same path as the same difference inside control industries absent the policy.

4.2 Why the industry-level policy timing is plausibly usable

- Table II shows that many pre-treatment industry characteristics do not predict reform: log variance of MRPK, number of firms, average firm capital, export share, state-owned sales share, concentration, and pre-existing FDI levels or growth are all basically unrelated to deregulation.

- This does not prove exogeneity, but it makes the common-trends assumption more plausible.
- The paper also stresses the political context: by the early 2000s there was a broader pro-FDI consensus, and the reforms were part of a push to attract foreign capital rather than a one-off crisis response like in 1991.

4.3 Pre-trends and robustness to time-varying shocks

- Figures 2–4 show no sharp pre-trends in average capital, in the relative outcomes of high-MRPK firms, or in separate high- and low-MRPK series.
- The dynamic treatment effects ramp up only gradually over several years, which is economically plausible in a setting with slow-moving capital adjustment and labor reallocation.
- The results are also robust to adding 2-digit industry-year fixed effects, 5-digit industry-year fixed effects, and state-year fixed effects.

4.4 Why endogenous foreign-investor targeting is not fatal

- The regressions do not use realized foreign-equity inflows as the treatment variable. They use the policy-induced change in whether an industry is open to foreign capital.
- So the estimates are still informative even if foreign investors choose particular firms endogenously inside treated industries.
- The authors explicitly note that foreign capital need not directly target the high-MRPK firms. It may instead fund other firms and free domestic credit to flow toward the constrained firms. What matters for misallocation is the *net* relative change in capital across firms.

4.5 Measurement error in MRPK

- Cross-sectional dispersion in $Revenue/K$ may mix together true wedges and measurement error.
- In the paper’s design, firm fixed effects absorb time-invariant firm-level measurement error, and classical measurement error in the left-hand-side outcomes does not bias point estimates.
- Misclassification of high-MRPK status due to noisy pre-treatment MRPK should mainly attenuate β_2 , since some truly constrained firms get misclassified as low-MRPK and vice versa.
- This is one reason the paper’s lower-bound aggregation procedure is valuable: it reduces the risk that noisy cross-sectional wedges mechanically inflate the implied aggregate gains.

5 Tables and figures

5.1 Table I: Summary statistics

Read:

- Treated and untreated firms look broadly comparable in age and scale, which is comforting for the difference-in-differences design.

- Mean MRPK is higher in treated industries before the reform (9.3 versus 7.5), consistent with the idea that the paper is studying sectors where capital-market frictions may have been meaningful.
- The sample is clearly the formal organized sector, not the full universe of tiny firms, which fits the paper's focus on aggregate productivity in treated formal industries.

TABLE I
SUMMARY STATISTICS.

	Control			Treated		
	Mean	S.D.	Median	Mean	S.D.	Median
Foreign (%)	4.1	19.7	0.0	7.0	25.6	0.0
State Owned (%)	3.7	18.8	0.0	2.0	14.1	0.0
Firm Age	26.4	18.9	21.0	26.7	18.9	21.0
Gross Fixed Assets	25.5	132.4	3.5	18.2	49.9	3.1
Sales/Revenues	64.3	274.4	12.3	57.8	146.6	11.9
Wages	3.6	19.7	0.6	4.2	9.8	0.8
MRPK (<i>Revenue/K</i>)	7.5	89.4	3.1	9.3	71.6	3.2
Observations	56,861			6288		

Note: This table reports summary statistics for the manufacturing firms appearing in the CMIE Prowess data set from 1995 to 2015. An observation is at the firm-year level. Firms' capital, income, salaries, and revenues are measured in millions of USD and are deflated with industry deflators.

TABLE II
ASSOCIATIONS BETWEEN INDUSTRY-LEVEL CHARACTERISTICS AND THE REFORM.

Dependent Variable	Reform = 1								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log (Variance in MRPK)	0.032 (0.027)								
Log (Num. Firms)		0.014 (0.025)							
Log (Avg. Firm Capital)			-0.033 (0.027)						
Share Export				0.116 (0.176)					
State-Owned Firms' Share of Total Sales					-0.006 (0.011)				
Herfindahl Index						-0.173 (0.123)			
lhs(Industry-Level FDI)							-0.029 (0.041)		
FDI Growth, 2001–2005								-0.012 (0.015)	
lhs(FDI 2005)-lhs(FDI 2001)									0.064 (0.052)
Observations	274	274	274	274	274	274	274	169	188

Note: This table reports the association of different industry-level characteristics with liberalization. Each observation is a 5-digit industry. In columns 1–7, the outcome of interest is whether an industry was reformed during the study period. In columns 8–9, the outcome is whether an industry was reformed in 2006, and we exclude industries deregulated in 2001. This is because foreign equity data are only available in Prowess from 2001, and therefore, the growth rate is not defined for the 2001 reform. Industry-level FDI (column 7) is the level in 2001. "lhs" in columns 7 and 9 corresponds to the inverse hyperbolic sine log transformation, defined as: $\log[X + (X^2 + 1)^{1/2}]$. Industries are weighted by size (measured as total capital in 2000, the last pre-treatment year).

5.2 Table II: Do pre-treatment industry characteristics predict reform?

Read:

- None of the obvious observables predicts liberalization in a meaningful way: the coefficient on log variance of MRPK is only 0.032, and the others are similarly small and insignificant.
- This is important because the paper's identifying assumption is about trends, not levels, but if observables were strongly related to selection into treatment it would weaken the design.
- The main takeaway is that treated industries do not appear to have been chosen because they already had unusually high measured misallocation or unusually strong FDI trends.

5.3 Figure 1 and Table III: First-stage evidence on foreign capital

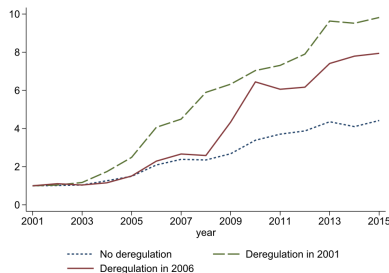


FIGURE 1.—Flow of foreign equities. *Note:* This figure plots the overall amount of foreign equity in Prowess for industries that have deregulated in 2001 (the red line), in 2006 (the green line), or whose regulation did not change during the period 1995–2015 (the blue line). The flows are normalized to 1 in 2001.

TABLE III
EFFECT OF THE REFORM ON PROXIES FOR FOREIGN DIRECT INVESTMENT.

Dependent Variable	Any Foreign Debt	Any Foreign Debt	Any Foreign Debt	Any Foreign Debt	Any Foreign Project	Any Foreign Project	Any Foreign Project	Any Foreign Project
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Reform_{it}$	0.030 (0.019)	-0.005 (0.013)	0.451 (0.345)	-0.078 (0.213)	0.004 (0.003)	0.0002 (0.003)	0.029 (0.022)	0.001 (0.020)
$Reform_{it} \times I_{it}^{High MRPK}$		0.064 (0.025)		0.963 (0.326)		0.007 (0.002)		0.050 (0.018)
<i>Fixed Effects</i>								
Firm	✓	✓	✓	✓	✓	✓	✓	✓
Firm Age	✓	✓	✓	✓	✓	✓	✓	✓
Size × Year	✓	✓	✓	✓	✓	✓	✓	✓
Observations	61,156	61,156	61,156	61,156	61,156	61,156	61,156	61,156

Note: This table reports the effect of the reform on access to foreign debt and the cost of capital projects financed by foreign investors. In columns 1–4, we use the universe of foreign loans since 2004 tracked by the RBI to study if firms issue any foreign debt (columns 1–2) or to estimate the total change in foreign debt (columns 3–4). Due to the large number of zeros, we use the inverse hyperbolic sine transformation (denoted $lhs(x)$) in columns 5–8. In columns 5–8, we use the data set CapEx, compiled by CMIE, which contains data on all large capital projects in India. The outcomes are an indicator variable for whether a firm is associated with a foreign-owned project (columns 5–6), and the lhs of the total spending on foreign-owned projects associated with the firm (columns 7–8). Standard errors are two-way clustered at the 4-digit industry and year level.

Read:

- Figure 1 shows the first-stage visually: foreign equity rises faster in industries liberalized in 2001 and 2006 than in industries with no deregulation during the sample window.
- Table III sharpens this at the firm level. High-MRPK firms are 6.4 percentage points more likely to access foreign debt and increase foreign-debt volumes by about 96%.
- The *CapEx* results point in the same direction: high-MRPK firms are more likely to undertake foreign-owned projects and spend more on them.
- The first stage is not perfect proof that the most constrained firms receive all the foreign capital directly, but it is strong evidence that liberalization actually changed financing access.

5.4 Table IV and Figure 2: Average effects of foreign-capital liberalization

TABLE IV
AVERAGE EFFECT OF THE FOREIGN CAPITAL LIBERALIZATION.

Dependent Variable	Revenues (1)	Capital (2)	Wages (3)	MRPK (4)
$Reform_{jt}$	0.108 (0.094)	0.318 (0.118)	0.159 (0.098)	-0.187 (0.088)
<i>Fixed Effects</i>				
Firm	✓	✓	✓	✓
Firm Age	✓	✓	✓	✓
Size × Year	✓	✓	✓	✓
Observations	58,391	60,096	59,162	57,017

Note: All dependent variables are in logs. $Reform_{jt}$ is an indicator variable equal to 1 if the industry had liberalized access to the international capital market in or before year t and zero otherwise. Size × Year are quartile fixed effects for firms' average pre-treatment capital interacted with year fixed effects. In column 4, MRPK is computed using $Revenue/K$ as a proxy for the marginal revenue product of capital. Standard errors are two-way clustered at the 4-digit industry and year level.

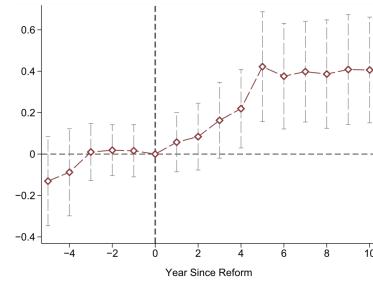


FIGURE 2.—Event study graph for the average effect of foreign capital liberalization on physical capital. Note: This figure reports the event study graph for the average effect of the liberalization on firms' physical capital. The dependent variable is in logs. The reform is normalized to take place in year 1. Each dot is the coefficient on the interaction between being observed t years after the reform and being in a treated industry. The confidence interval is at the 95% level.

Read:

- On average, treated industries see more capital. The capital coefficient is 0.318, while average MRPK falls by 0.187.
- The revenue and wage coefficients are positive but imprecise, which already hints that average effects may hide important heterogeneity.
- Figure 2 shows no obvious pre-trend and a gradual buildup after treatment, with the capital effect stabilizing only about five years later.
- Economically, this says liberalization raises capital overall; the next question is whether that extra capital goes to the right firms.

5.5 Table V: Heterogeneous effects by ex ante MRPK

Read:

- This is the central table. Relative to low-MRPK firms, high-MRPK firms increase revenues by 22.6%, capital by 52.7%, and wages by 28.0%, while their MRPK falls by 32.6%.
- Low-MRPK firms are basically unaffected: the coefficients on $Reform_{jt}$ alone are close to zero.
- Since the high-MRPK firms began with more than 160% higher MRPK than the low-MRPK firms, the 33% decline does not just reflect mean reversion; it reflects a genuine narrowing of within-industry dispersion.

TABLE V
HETEROGENEOUS EFFECTS OF FOREIGN CAPITAL LIBERALIZATION BY FIRMS' EX ANTE MRPK.

Dependent Variable	Revenues (1)	Capital (2)	Wages (3)	MRPK (4)
$Reform_{jt} \times I_t^{HighMRPK}$	0.226 (0.076)	0.527 (0.075)	0.280 (0.058)	-0.326 (0.107)
$Reform_{jt}$	-0.024 (0.125)	0.017 (0.089)	-0.002 (0.097)	0.004 (0.102)
<i>Fixed Effects</i>				
Firm	✓	✓	✓	✓
Firm Age	✓	✓	✓	✓
Size $\times$ Year	✓	✓	✓	✓
Observations	58,391	60,096	59,162	57,017

Note: All dependent variables are in logs. Firms are classified as high MRPK if their average MRPK in the pre-treatment period from 1995–2000 is above the 4-digit industry median. MRPK is estimated with the $Revenue/K$ method. Size $\times$ Year are quartile fixed effects for firms' average pre-treatment capital interacted with year fixed effects. Standard errors are two-way clustered at the 4-digit industry and year level.

- This is the paper's cleanest micro evidence that capital misallocation falls.

5.6 Figures 3 and 4: Dynamics and pre-trends

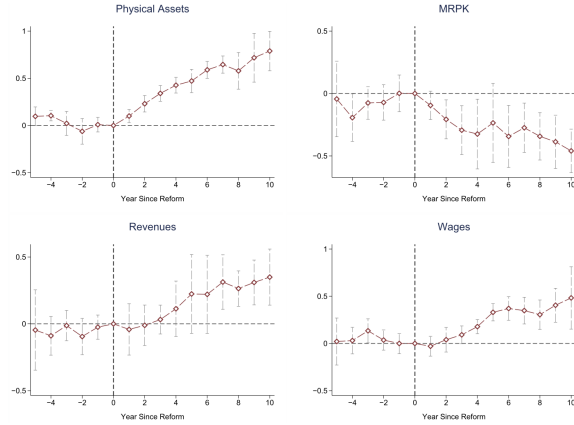


FIGURE 3.—Event study graphs for the relative effect of foreign capital liberalization on high MRPK firms. *Note:* This figure reports event study graphs for the relative effects of the liberalization on firms with high pre-treatment MRPK relative to those with low pre-treatment MRPK in treated industries relative to untreated industries. The reform is normalized to take place in year 1. Each dot is the coefficient on the interaction between being observed t years after the reform and being a high MRPK firm in a treated industry. All dependent variables are in logs. The confidence intervals are at the 95% level.

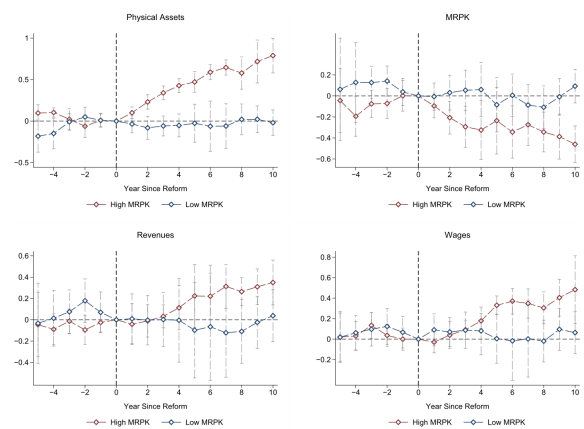


FIGURE 4.—Separate event studies for high and low MRPK firms. *Note:* This figure reports the effect of FDI deregulation for high and low MRPK firms separately for physical assets, MRPK, revenues, and the wage bill. The dependent variables are in logs. The reform is normalized to take place in year 1. Each dot is the coefficient on the interaction between being observed t years after the reform and being in a treated industry. The confidence interval is at the 95% level.

Read:

- Figure 3 shows the dynamic relative treatment effects for high-MRPK firms. Before the reform, the lines are flat; after the reform, capital rises steadily and MRPK falls steadily.
- The effects are gradual. By year 10 after the reform, the relative capital effect is roughly 79% and the relative MRPK effect is about -46%.
- Figure 4 makes the same point in a different way: the treatment effect is driven almost entirely by the high-MRPK firms, while low-MRPK firms remain close to zero throughout.
- These figures are critical because they support the common-trends assumption and show that the reallocation mechanism strengthens over time rather than appearing only as a one-year blip.

5.7 Table VI: Industry-level capital and MRPK dispersion in *Prowess* and the *ASI*

Read:

- In *Prowess*, treated 4-digit industries increase total capital by 27.5% and reduce the variance of MRPK by roughly 40% to 68%, depending on the specification.
- In the representative *ASI* data, total capital rises by 61.6% and the variance of MRPK falls by roughly 77% to 78%.
- This matters because the core micro data are not fully representative. Table VI shows that the main message survives in a more representative establishment dataset.

Dependent Variable	Total Capital (1)	Variance(MRPK)	
		(2)	(3)
Share of (4-digit) Industry Treated	0.275 (0.145)	<i>Panel A: Prowess</i> −0.401 (0.168)	−0.684 (0.320)
<i>Fixed Effects</i>			
Industry (4-digit)	✓	✓	✓
Industry (2-digit) × Year	✓	✓	✓
<i>Controls</i>			
Nb firms	—	—	✓
Observations	2278	1789	1789
Share of (4-digit) Industry Treated	0.616 (0.270)	<i>Panel B: ASI</i> −0.783 (0.327)	−0.771 (0.321)
<i>Fixed Effects</i>			
Industry (4-digit)	✓	✓	✓
Industry (2-digit) × Year	✓	✓	✓
<i>Controls</i>			
Nb firms	—	—	✓
Observations	2184	2120	2120

Note: This table reports the effects of the reform on log total 4-digit industry-level capital and the 4-digit industry-level log variance of MRPK in Prowess (Panel A) and the ASI (Panel B). 'Nb firms' refers to a control for the log number of firms in an industry-year. Standard errors are two-way clustered at the 4-digit industry and year level.

TABLE VII
EFFECT OF FOREIGN CAPITAL LIBERALIZATION ON TFPQ.

Dependent Variable	TFPQ (1)	TFPQ (2)
$Reform_{jt}$	0.157 (0.166)	0.106 (0.143)
$Reform_{jt} \times I_i^{High MRPK}$		0.084 (0.074)
<i>Fixed Effects</i>		
Firm	✓	✓
Firm Age	✓	✓
Size × Year	✓	✓
Observations	43,791	43,791

Note: All dependent variables are in logs. Firms are classified as high MRPK if their average MRPK in the pre-treatment period from 1995 to 2000 is above the 4-digit industry median. MRPK is estimated with the *Revenue/K* method. Size × Year are quartile fixed effects for firms' average pre-treatment capital interacted with year fixed effects. TFPQ is measured by estimating revenue production functions using the methodology of Levinsohn and Petrin (2003) and subtracting log average price from log TFPR. Standard errors are two-way clustered at the 4-digit industry and year level.

5.8 Table VII: Within-firm productivity

Read:

- The TFPQ coefficients are positive but imprecise: 0.157 on average and 0.084 for the interaction with high MRPK.
- So the paper does *not* claim strong within-firm productivity gains from the reform.
- That is conceptually useful. The aggregate gains the paper estimates mainly come from better allocation, not from a large measured rise in firms' own physical productivity.

5.9 Table VIII and Figure 5: Heterogeneity by local financial development

Read:

- The triple interaction with local financial development is negative for capital (−0.233) and wages (−0.189), and positive for MRPK (0.138).
- That means the reform has bigger reallocation effects in places where the local banking sector was weaker before the reform.

TABLE VIII
HETEROGENEITY BY LOCAL FINANCIAL DEVELOPMENT.

Dependent Variable	Revenues (1)	Capital (2)	Wages (3)	MRPK (4)
$Reform_{jt} \times I_i^{HighMRPK} \times Financial\ Development_t$	-0.095 (0.078)	-0.233 (0.084)	-0.189 (0.060)	0.138 (0.047)
$Reform_{jt} \times I_i^{HighMRPK}$	0.203 (0.079)	0.527 (0.092)	0.259 (0.064)	-0.349 (0.121)
<i>Fixed Effects</i>				
Firm	✓	✓	✓	✓
Firm Age	✓	✓	✓	✓
Size × Year	✓	✓	✓	✓
Observations	53,109	54,692	53,852	51,873

Note: All dependent variables are in logs. $Reform_{jt}$ is an indicator variable equal to 1 if the industry has liberalized access to the international capital market. Firms are classified as high MRPK if their average MRPK in the pre-treatment period from 1995 to 2000 is above the 4-digit industry median. MRPK is calculated using the *Revenue/K* method. Size × Year are quartile fixed effects for firms' average pre-treatment capital interacted with year fixed effects. Local financial development is provided using the log average amount of bank credit in the state in the pre-treatment period. All double and single interactions of the triple-differences specification are included in the regressions. Standard errors are two-way clustered at the 4-digit industry and year level.

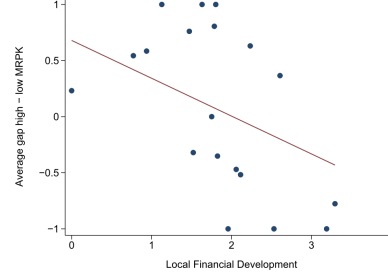


FIGURE 5.—Ex ante MRPK dispersion by state-level financial development. Note: This figure plots a binscatter plot of a measure of the pre-treatment dispersion of MRPK by state-level financial development. For each state-industry bin, we compute the average MRPK for high and low MRPK firms and take the difference. We then express these differences as a fraction of the total distance (namely, we compute: $[(MRPK_{high} - MRPK_{low}) / (MRPK_{high} + MRPK_{low})]$) and plot this measure against state-level financial development.

- The paper translates these coefficients into a very intuitive comparison: high-MRPK firms in a state at the 25th percentile of bank credit see MRPK fall by about 45%, versus only 16% at the 75th percentile.
- Figure 5 reinforces the mechanism visually: ex ante MRPK dispersion is larger in states with lower local financial development.

5.10 Table IX: Product-market outcomes

Read:

- Product prices fall after liberalization. The average price effect is -17.1% , and high-MRPK firms see an additional -12.3% effect, implying roughly a 21% total price decline for those firms.
- Output rises too: the average effect is 26.8% , with an extra 24.2% for high-MRPK firms.
- The product portfolio expands at the top. High-MRPK firms increase the number of products by about 3.4% and are almost 10 percentage points more likely to add products.
- This is consistent with constrained firms using the new financing access to scale, cut prices, and broaden their product scope.

TABLE IX
EFFECT OF FOREIGN CAPITAL LIBERALIZATION ON PRODUCT OUTCOMES.

Dependent Variable	Price		Output		Log(# Products)	Pr(Addition)	Pr(Deletion)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$Reform_{jt}$	-0.171 (0.031)	-0.084 (0.053)	0.268 (0.058)	0.097 (0.087)	0.004 (0.025)	-0.075 (0.016)	-0.005 (0.035)
$Reform_{jt} \times I_i^{HighMRPK}$		-0.123 (0.065)		0.242 (0.125)	0.034 (0.014)	0.097 (0.020)	-0.051 (0.048)
<i>Fixed Effects</i>							
Firm	✓	✓	✓	✓	✓	✓	✓
Firm Age	✓	✓	✓	✓	✓	✓	✓
Size × Year	✓	✓	✓	✓	✓	✓	✓
Firm × Product	✓	✓	✓	✓	—	—	—
Observations	103,035	103,035	103,974	103,974	32,660	32,660	32,660

Note: In columns 1–4, each observation is at the firm-product-year level. In columns 5–7, each observation is at the firm-year level. Firms are classified as high MRPK if their average MRPK in the pre-treatment period from 1995 to 2000 is above the 4-digit industry median. MRPK is calculated using the *Revenue/K* method. Size × Year are quartile fixed effects for firms' average pre-treatment capital interacted with year fixed effects. Standard errors are two-way clustered at the 4-digit industry and year level.

TABLE X
EFFECT OF FOREIGN CAPITAL LIBERALIZATION BY FIRMS' EX ANTE MRPL.

Dependent Variable	Revenues (1)	Capital (2)	Wages (3)	MRPL (4)
$Reform_{jt} \times I_i^{HighMRPL}$	-0.039 (0.086)	0.198 (0.119)	0.237 (0.069)	-0.276 (0.061)
$Reform_{jt}$	0.120 (0.105)	0.121 (0.072)	0.024 (0.104)	0.096 (0.096)
<i>Fixed Effects</i>				
Firm	✓	✓	✓	✓
Firm Age	✓	✓	✓	✓
Size × Year	✓	✓	✓	✓
Observations	43,407	43,384	43,407	43,407

Note: All dependent variables are in logs. High MRPL firms are defined in an analogous way to high MRPK firms using the *Revenue/L* method. $Reform_{jt}$ is an indicator variable equal to 1 if the industry has liberalized access to the foreign capital market. Size × Year are quartile fixed effects for firms' average pre-treatment capital interacted with year fixed effects. Standard errors are two-way clustered at the 4-digit industry and year level.

5.11 Table X: Extension to labor misallocation

Read:

TABLE XI
EFFECTS OF FOREIGN CAPITAL MARKET LIBERALIZATION ON THE SOLOW
RESIDUAL OF TREATED INDUSTRIES.

	Increase in Solow Residual
Lower Bound	3.4%
Conventional Approach	16.3%
Lower Bound Allowing for Cumulative Effects	6.2%
Non-Linear Approximation	6.0%

Note: This table reports the estimates of the effect of the foreign capital liberalizations in 2001 and 2006 on treated industries' Solow residual. Rows 1, 2, and 3 use a first-order approximation (equation (3)). The last row uses a non-linear approximation, described in Appendix C of the Supplemental Material. The estimates are generated using the Prowess data set. Rows 1, 3, and 4 identify the wedges following the lower bound approach discussed above. Row 2 uses cross-sectional data to identify the baseline wedges.

- High-MRPL firms increase wage bills by 23.7% relative to low-MRPL firms, and their MRPL falls by 27.6%.
- So the reform appears to reduce labor misallocation as well, not just capital misallocation.
- This is important because financing frictions can distort employment too when firms must finance payroll and hiring adjustment is not frictionless.

5.12 Table XI: Aggregate effect on the Solow residual

Read:

- The lower-bound first-order estimate is that the policy raises the Solow residual of treated industries by 3.4%.
- Using conventional cross-sectional wedge estimates gives a much larger 16.3%, which the authors treat as less credible because it is more exposed to measurement-error inflation.
- Allowing for cumulative treatment effects over time pushes the lower bound to 6.2%, and the non-linear approximation gives a very similar 6.0%.
- This table closes the paper: the micro reallocation effects are not just statistically interesting; they are large enough to matter for aggregate productivity.

6 Short summary

- **Conclusion.** Foreign equity liberalization in India reduced within-industry capital and labor misallocation by allowing initially constrained firms to expand.
- **Intuition.** The key margin is not simply more capital, but better allocation of capital toward firms with high marginal revenue products.
- **Empirical lesson.** A natural experiment can identify changes in wedges more credibly than cross-sectional dispersion measures and can be used to bound aggregate productivity gains.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Liberalizing access to foreign capital reallocates resources toward firms that had been most constrained before the reform.
- **Micro magnitude.** The headline relative effects for high-MRPK firms are large: roughly 23% more revenue, 53% more capital, 28% more wages, and 33% lower MRPK.
- **Mechanism evidence.** The effects are strongest where local banking systems were weak, suggesting that domestic credit frictions were an important source of the original misallocation.
- **Product-market implication.** Constrained firms do not just buy more capital; they also lower prices, expand output, and add products.
- **Aggregate implication.** Even the conservative lower-bound exercise implies economically meaningful aggregate productivity gains for treated industries, on the order of 3% to 6%.
- **Methodological contribution.** The paper offers a disciplined way to move from reduced-form policy variation to aggregate productivity statements without relying entirely on cross-sectional wedge dispersion.

7.2 Economic intuition

- When domestic credit markets allocate capital poorly, firms with high marginal returns stay too small. Opening access to foreign equity effectively loosens that bottleneck.
- The average capital stock can rise without improving efficiency, but efficiency rises when the additional capital disproportionately flows to firms with the highest wedges.
- The banking-sector heterogeneity suggests that foreign capital acts partly as a substitute for weak domestic intermediation.
- The slow dynamics in the event studies are economically sensible: capital adjustment, hiring, and product expansion all take time.

7.3 Takeaway

This paper is one of the cleanest empirical demonstrations that a concrete policy can reduce misallocation inside a developing-country corporate sector. Its central message is that foreign capital market integration can improve not just the *level* of capital, but also its *allocation*, especially when domestic banking systems are weak. Just as importantly, the paper shows how to translate that micro reallocation into a conservative aggregate productivity estimate, which is exactly the bridge the misallocation literature has often lacked.